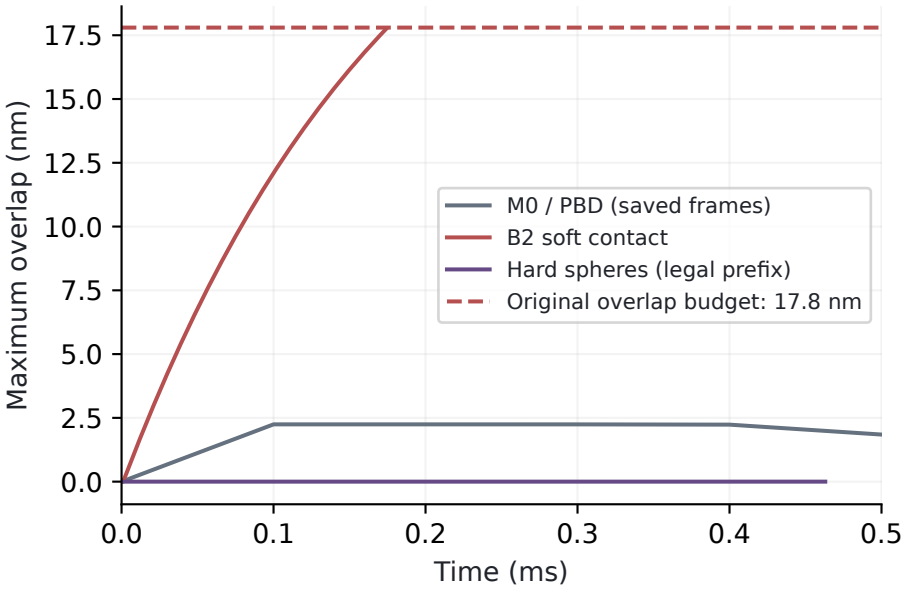
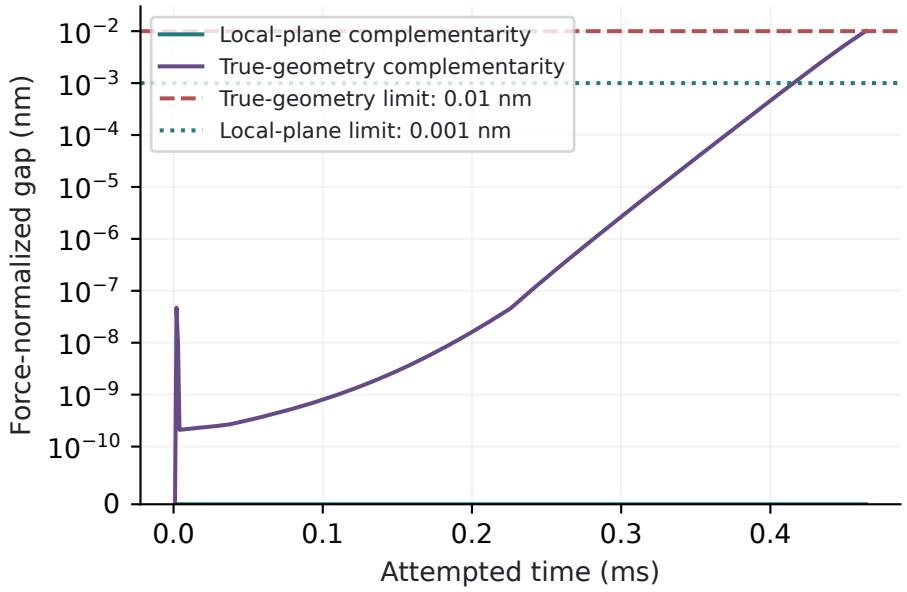
Hard-sphere contact: geometry, complementarity and cost

a Actual overlap



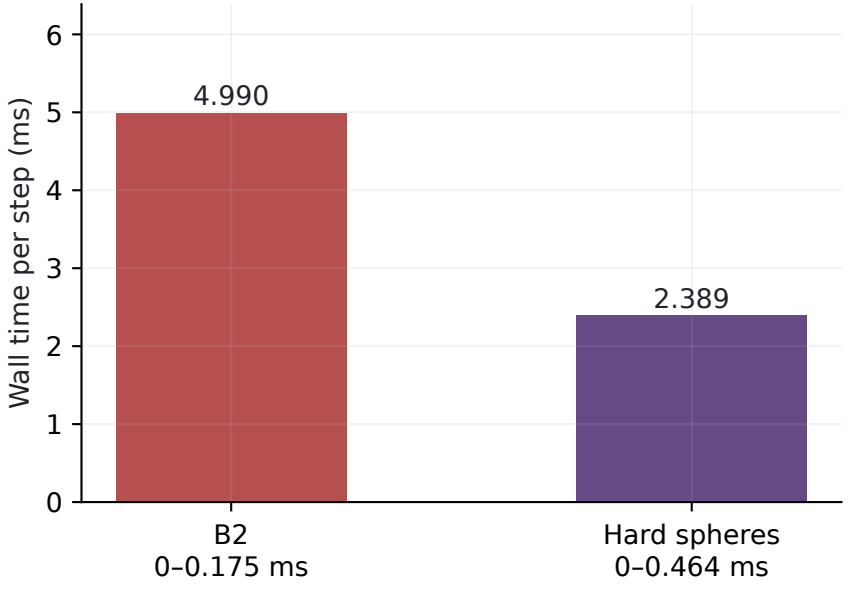
b Independent reaction-gap checks



c Two- and four-sphere independent references

Reference case	Gate	Error (nm)
separating	PASS	0
compressing	PASS	0
release	PASS	0
no contact	PASS	0
four simultaneous	PASS	0
four network translation	PASS	0

d Measured whole-step cost



Different valid windows; no matched-time speedup claim

EXPLORATORY | HOMOGENEOUS_WATER_CONTROL_PROXY | PHYSICS_UNCALIBRATED | training_ready=false